

Centralized Authentication & SSO System Documentation

Project Overview

This project implements a centralized authentication system with Single Sign-On (SSO) using Keycloak. Users log in once and access multiple applications without re-authentication.

Architecture Overview

The system includes React App, SvelteKit App, and Keycloak Identity Provider running in Docker.

Authentication Flow

User is redirected to Keycloak for login. After authentication, a session is created and shared across all apps.

Keycloak Configuration

Two clients are configured: react-app and svelte-app with respective redirect URIs.

SSO Behavior

Login in one app automatically authenticates the other due to shared Keycloak session.

Security Model

Uses OAuth2 Authorization Code Flow with PKCE and secure token handling.

The system avoids custom password and session management code. Instead, it delegates authentication to Keycloak, reducing security risk and ensuring the solution follows established OAuth2 and OpenID Connect standards.

Conclusion

This system demonstrates centralized authentication and scalable SSO architecture using Keycloak.